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RELIABILITY TEST METHOD FOR PRINTED WIRING BOARD SUBSTRATE

Abstract

A reliability test method for a printed wiring board substrate includes measuring a load at which a crack occurs in the printed wiring board substrate using a nanoindenter.

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Background/Summary

TECHNICAL FIELD

[0001] The present embodiment relates to a reliability test method for a printed wiring board substrate.

BACKGROUND ART

[0002] In recent years, in electronic devices such as computers, the capacity and speed of signals used have been increased, and semiconductor packages used in these electronic devices have also been highly integrated and functionalized.

[0003] In addition, semiconductor packages are required to be adapted to various temperature conditions depending on devices to be used, installation environments, and the like, and ensuring long-term reliability under temperature fluctuations is becoming increasingly important.

[0004] As a printed wiring board substrate, a substrate containing a resin component as an insulating material is used. However, due to a difference in thermal expansion coefficient between the resin component and an inorganic component such as a conductor layer or a semiconductor chip, stress may be generated when the temperature changes, and a crack may occur in the printed wiring board substrate. In particular, since the number of build-up layers tends to increase, the stress caused by the difference in thermal expansion coefficient is becoming large, and the problem of cracks occurring due to temperature changes has become prominent, in recent years.

[0005] As a reliability test method for a printed wiring board substrate, a temperature cycle test is generally performed (for example, see Patent Literature 1).

CITATION LIST

Patent Literature

[0006] Patent Literature 1: JP 2006-093618 A

SUMMARY OF INVENTION

Technical Problem

[0007] The temperature cycle test is an effective test method for confirming long-term reliability of a test object with respect to a temperature change by repeating heating and cooling. On the other hand, since the temperature cycle test requires a period of several weeks to several months to obtain a test result, a simpler reliability evaluation method is desired.

[0008] In view of the above-described circumstances, a problem of the present embodiment is to provide a reliability test method for a printed wiring board substrate that can be easily performed.

Solution to Problem

[0009] As a result of diligent studies to solve the above problem, the present inventors have found that the above problem can be solved by the following embodiment.

[0010] That is, the present embodiment relates to the following items [1] to [6].

[0011] [1] A reliability test method for a printed wiring board substrate, the method including [0012] measuring a load at which a crack occurs in the printed wiring board substrate, using a nanoindenter.

[0013] [2] The reliability test method for a printed wiring board substrate according to item [1] above, in which the printed wiring board substrate includes a resin cured product and a glass cloth.

[0014] [3] The reliability test method for a printed wiring board substrate according to item [1] or [2] above, in which the printed wiring board substrate has two or more layers each including a resin cured product and a glass cloth.

[0015] [4] The reliability test method for a printed wiring board substrate according to item [2] above, in which an indenter of the nanoindenter is pushed into the resin cured product.

[0016] [5] The reliability test method for a printed wiring board substrate according to any of items [1] to [4] above, in which the indenter of the nanoindenter is pushed into a cross-sectional surface

of the printed wiring board substrate.

[0017] [6] The reliability test method for a printed wiring board substrate according to any of items [1] to [5] above, in which the indenter of the nanoindenter is a Berkovich indenter.

Advantageous Effects of Invention

[0018] According to the present embodiment, it is possible to provide a reliability test method for a printed wiring board substrate that can be easily performed.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0019] FIG. 1 is a graph for explaining a load-displacement curve.

[0020] FIG. 2 is a graph for explaining a load-contact stiffness curve.

[0021] FIG. 3 is a schematic view for explaining positions observed by an optical microscope in a liquid bath temperature cycle test.

[0022] FIG. 4 is a schematic view for explaining a position where a test piece is sampled and a test region for a fracture test by a nanoindenter.

[0023] FIG. 5 is a diagram for explaining a method of analyzing a load-contact stiffness curve.

[0024] FIG. 6 is a graph representing a correlation between crack occurrence loads W1 and crack occurrence rates after temperature cycle tests (1,000 cycles), made based on results of Examples.

DESCRIPTION OF EMBODIMENTS

[0025] The action mechanism described in the present specification is inferential and does not limit the mechanism of providing an effect of the present embodiment.

[0026] An aspect in which matters described in the present specification are arbitrarily combined is also included in the present embodiment.

Reliability Test Method for Printed Wiring Board Substrate

[0027] A reliability test method for a printed wiring board substrate according to the present embodiment measures a load at which a crack occurs in the printed wiring board substrate using a nanoindenter.

[0028] According to the reliability test method of the present embodiment, since a long period of time is not required for the test unlike a temperature cycle test, the reliability of the printed wiring board substrate can be evaluated in a simple manner.

Nanoindenter

[0029] The nanoindenter used in the reliability test method of the present embodiment is a microhardness tester that pushes a minute indenter into a surface of a sample to a depth ranging from a few nanometers to several tens of micrometers, and can measure the relationship between the displacement amount and the load at that time.

Printed Wiring Board Substrate

[0030] The printed wiring board substrate, which is a test object in the reliability test method of the present embodiment, is not particularly limited, and a known printed wiring board substrate can be used as a test object.

[0031] An insulating layer included in the printed wiring board substrate may be a layer that contains a resin and contains no glass cloth, or may be a layer that contains a resin and a glass cloth. A layer including a resin cured product and a glass cloth is preferable as the layer that contains a resin and a glass cloth. In the present specification, the “resin cured product” means a cured product that includes at least a resin and includes no glass cloth, and is preferably a cured product of a thermosetting resin composition containing a thermosetting resin.

[0032] The thermosetting resin composition may contain, as a component other than the thermosetting resin, a curing agent, a curing accelerator, an inorganic filler, and the like, if needed.

[0033] The insulating layer included in the printed wiring board substrate may be a single layer or

multiple layers.

[0034] A specific example of the test object for the reliability test method of the present embodiment is, for example, a printed wiring board substrate having two or more layers each including a resin cured product and a glass cloth.

[0035] The printed wiring board substrate may have a member other than the insulating layer which may be generally included in the printed wiring board substrate. Examples of the member other than the insulating layer include a conductor layer formed on a surface of the substrate, a conductor layer formed between insulating layers, and a semiconductor chip mounted on a surface of the substrate.

[0036] The content of the resin cured product in the layer containing the resin cured product and the glass cloth is not particularly limited, and may be 25% to 75% by mass, may be 30% to 70% by mass, and may be 35% to 65% by mass, for example.

[0037] The thickness per layer of the layer containing the resin cured product and the glass cloth is not particularly limited, and may be 200 to 1,900 μm , may be 500 to 1,800 μm , and may be 800 to 1,700 μm , for example.

[0038] When the substrate has two or more layers each containing the resin cured product and the glass cloth, the number of layers each containing the resin cured product and the glass cloth is not particularly limited, and may be 2 to 19 layers, may be 5 to 18 layers, and may be 8 to 17 layers, for example.

Test Conditions

[0039] Next, test conditions of a fracture test using the nanoindenter will be described.

[0040] In the following description, a test for measuring a load at which a crack occurs in a printed wiring board substrate using a nanoindenter is referred to as the “fracture test”.

[0041] In the following description, a load at which a crack occurs in a printed wiring board substrate measured in the fracture test is referred to as a “crack occurrence load”.

[0042] Although the position of the printed wiring board substrate subjected to the fracture test by the nanoindenter is not particularly limited, it is preferable to select a site where cracks are likely to occur in a temperature cycle test, for example. The site where cracks are likely to occur in a temperature cycle test is a site where stress is likely to be generated in the printed wiring board substrate. Therefore, the crack occurrence load is easily obtained in the range of the fracture test by the nanoindenter.

[0043] In a plan view of the printed wiring board substrate, an example of the site where the fracture test is performed is preferably an end portion of the printed wiring board substrate from the viewpoint of easily obtaining the crack occurrence load. Further, from the viewpoint of easily obtaining the crack occurrence load, the indenter of the nanoindenter is preferably pushed into a cross sectional surface of the printed wiring board substrate. Since the printed wiring board substrate is obtained by cutting out the outer shape thereof in a plan view to have a desired shape, an end surface of the printed wiring board substrate corresponds to the cross sectional surface.

[0044] In a cross sectional view of the printed wiring board substrate, an example of a site where the fracture test is performed is substantially a central portion of the printed wiring board substrate in the thickness direction.

[0045] That is, the site where the fracture test is performed is preferably a cross sectional surface of an end portion of the printed wiring board substrate in a plan view, and is preferably substantially a central portion of the printed wiring board substrate in a cross sectional view.

[0046] In performing the fracture test, it is preferable to process the printed wiring board substrate by cutting or the like to form a test piece including a region to be subjected to the fracture test, in order to mount same on the nanoindenter.

[0047] The surface of the test piece into which the indenter of the nanoindenter is pushed is preferably smoothed in advance by polishing before or after being sampled as a test piece, from the viewpoint of suppressing occurrence of variation in test results due to unevenness of the surface.

Although the polishing method is not particularly limited, mechanical polishing or mechanochemical polishing is preferable.

[0048] If necessary, the test piece sampled from the printed wiring board substrate may be subjected to a pretreatment before the fracture test. Examples of the pretreatment include heat treatment and cooling treatment. By performing heat treatment and cooling treatment as the pretreatment, internal stress is accumulated in the test piece, and the crack occurrence load is easily obtained in the range of the fracture test by the nanoindenter.

Conditions of Fracture Test

[0049] Although the type of the indenter of the nanoindenter is not particularly limited, a Berkovich indenter is preferable from the viewpoint of versatility. The Berkovich indenter is an indenter in which the shape of a surface pushed into the test piece is a triangular pyramid (with a ridge interval of 115°). The type and the ridge interval of the indenter are not particularly limited, and a Berkovich indenter (a ridge interval of 100°), a Rockwell indenter, a Vickers indenter, a Knoop indenter, or a Shore indenter may be used.

[0050] Since the indenter of the nanoindenter is minute, when a substrate including a plurality of kinds of materials such as a layer containing a resin cured product and a glass cloth is to be tested, the obtained load-displacement curve may change depending on the position where the indenter is pushed into.

[0051] In a case where the object to be measured is a substrate having a layer containing a resin cured product and a glass cloth, when the indenter is pushed into the glass cloth or into the resin cured product at a position close to the glass cloth, a crack of the glass cloth is likely to occur. Meanwhile, when the indenter is pushed into the resin cured product at a position appropriately distant from the glass cloth, a crack at an interface between the glass cloth and the resin cured product and a crack of the resin cured product are likely to occur, and when the indenter is pushed into the resin cured product at a position further distant from the glass cloth, only a crack of the resin cured product is likely to occur.

[0052] In a temperature cycle test of the substrate having the layer containing the resin cured product and the glass cloth, since cracks at the interface between the resin cured product and the glass cloth and cracks of the resin cured product are likely to become a problem, it is preferable to adjust the position where the indenter is pushed into to a position where a crack at the interface between the resin cured product and the glass cloth and a crack of the resin cured product are likely to occur.

[0053] In view of the above, the indenter is preferably pushed into the resin cured product, and is preferably pushed into the resin cured product at a position appropriately distant from the glass cloth. On the surface into which the indenter is pushed, the shortest distance between the glass cloth and the center position on the resin cured product where the indenter is pushed into may be 1 to 10 μm , may be 2 to 8 μm , and may be 3 to 7 μm , for example.

[0054] The position where the indenter is pressed against can be adjusted by an optical microscope and a micrometer attached to the nanoindenter.

[0055] The indentation speed of the indenter is not particularly limited, but may be 15 to 55 mN/s, may be 25 to 50 mN/s, and may be 35 to 45 mN/s, for example.

[0056] The temperature at which the fracture test is performed is not particularly limited, but may be 5°C. to 50°C. , may be 10°C. to 40°C. , and may be 20°C. to 30°C. from the viewpoint of workability.

[0057] Although the atmosphere in which the fracture test is performed is not particularly limited, the fracture test is preferably performed in air from the viewpoint of workability.

Analysis Conditions

[0058] A load-displacement curve is obtained by the fracture test described above.

[0059] One example of the load-displacement curve is shown in FIG. 1.

[0060] As in the load-displacement curve shown in FIG. 1, the load as a whole tends to increase as

the indentation depth of the indenter into the test piece, that is, the displacement increases; however, a region in which the increase of the load becomes gentle is generated as in the region A shown in FIG. 1 due to a crack of the test piece.

[0061] FIG. 2 shows a curve (hereinafter, also referred to as the “load-contact stiffness curve”) obtained by plotting the contact stiffness, which is calculated as the slope of the load-displacement curve shown in FIG. 1, on the vertical axis and plotting the load on the horizontal axis. As shown in FIG. 2, the region A in FIG. 1 appears as a downward peak at which the contact stiffness decreases. Hereinafter, this peak is referred to as the “crack occurrence peak”.

[0062] As described later in Examples, the load at which the crack occurrence peak appears correlates with the crack occurrence rate in a temperature cycle test. Therefore, the reliability of the printed wiring board substrate can be evaluated by measuring the load (hereinafter also referred to as the “crack occurrence load”) at which the crack occurrence peak appears. For example, by grasping the correlation between the crack occurrence load and the crack occurrence rate in the temperature cycle test in advance, the crack occurrence rate in the temperature cycle test can be predicted from the crack occurrence load.

[0063] When a plurality of crack occurrence peaks appears, a threshold value may be set for the magnitude of the crack occurrence peak, and a crack occurrence peak with magnitude equal to or greater than a predetermined value may be used as an index for reliability evaluation. The threshold value of the magnitude of the crack occurrence peak may be, for example, a contact stiffness decrease rate (%) obtained from the following equation (1).

Contact stiffness decrease rate (%) = $(S1 - S2) \times 100 / S1$ (1):

[0064] In the equation, S1 means the contact stiffness obtained when the contact stiffness starts to decrease in the crack occurrence peak, and S2 means the contact stiffness at the peak top of the crack occurrence peak.

[0065] Incidentally, the “contact stiffness obtained when the contact stiffness starts to decrease” can be determined by the method described in Examples, for example. In addition, the “peak top” means the top of the crack occurrence peak (the point representing the minimum value of the contact stiffness in the peak).

[0066] After the crack occurrence peak used as an index for the reliability evaluation is determined, a load that gives a peak top of the crack occurrence peak may be used for the reliability evaluation, for example.

[0067] In a case where a plurality of crack occurrence peaks satisfying the above threshold value appears, it is preferable to use the crack occurrence peak appearing at the lowest load for the reliability evaluation among the peaks.

[0068] The crack occurrence peak suitable as an index for reliability may vary also depending on the position where the indenter is pushed into. Therefore, it is preferable that the position where the indenter is pushed into be determined, a temperature cycle test and the fracture test by the nanoindenter be performed on a plurality of types of test objects in advance under the condition under which the indenter is pushed into the position, and a crack occurrence peak having a high correlation with the temperature cycle test be grasped among the crack occurrence peaks appearing.

EXAMPLES

[0069] Hereinafter, the present embodiment will be specifically described by providing examples. However, the present embodiment is not limited to the following examples.

Preparation of Printed Wiring Board Substrate

Production Examples 1 to 6

[0070] Printed wiring board substrates 1 to 6 (substrates each having a thickness of 1.7 to 2.0 mm, with the outer shape being cut out to have a rectangular shape of 60 mm×60 mm in a plan view) having the following multilayer structure were produced using a build-up material (manufactured by Ajinomoto Co., Inc., product name: “GX92,” thickness: 30 μm) (hereinafter also referred to as

“Bu”) and copper foil (thickness: 18 μm or 12 μm), with a laminated plate obtained by lamination-molding twelve sheets of a predetermined insulating layer forming material (prepreg containing a thermosetting resin composition and a glass cloth) as a core layer. The respective core layers of obtained printed wiring board substrates **1** to **6** are different from one another.

Substrate Configurations

[0071] Copper foil (18 μm)/Bu (30 μm)/Bu (30 μm)/copper foil (18 μm)/Bu (30 μm)/Bu (30 μm)/copper foil (18 μm)/Bu (30 μm)/Bu (30 μm)/copper foil (18 μm)/Bu (30 μm)/Bu (30 μm)/copper foil (12 μm)/core layer (1150 to 1350 μm)/copper foil (12 μm)/Bu (30 μm)/Bu (30 μm)/copper foil (18 μm)/Bu (30 μm)/Bu (30 μm)/copper foil (18 μm)/Bu (30 μm)/Bu (30 μm)/copper foil (18 μm)/Bu (30 μm)/Bu (30 μm)/copper foil (18 μm)

[0072] The thicknesses in the parentheses mean the thicknesses of the respective layers before lamination.

[0073] Thereafter, a silicon semiconductor chip (rectangular shape having an outer shape of 20 mm $\times$ 20 mm in a plan view and a thickness of 0.775 mm) was disposed at the center of one surface of each of the printed wiring board substrates obtained above so that the circuit surface faced downward, and the gap between the semiconductor chip and the substrate was sealed using a liquid sealing material (Resonac Holdings Corporation, product name “CEL-C-3730 series”), thereby obtaining semiconductor chip mounting substrates **1** to **6** for a liquid bath temperature cycle test.

Liquid Bath Temperature Cycle Test

[0074] In the liquid bath temperature cycle test, five substrates were prepared for each of the semiconductor chip mounting substrates, and the five substrates were subjected to a temperature cycle test in which heating and cooling were repeated under the following conditions: at -65°C . for 5 minutes, at 150°C . for 5 minutes, and with a transfer time between a low-temperature bath and a high-temperature bath of 18 seconds. The substrates were taken out at the point when 1,000 cycles had been completed, and a predetermined position of each of the substrates was observed by an optical microscope to confirm the presence or absence of occurrence of cracks.

[0075] FIG. **3** shows a transparent perspective view for explaining positions observed by an optical microscope. In a substrate **1** shown in FIG. **3**, the four corners of the substrate indicated by regions **10** (hatched portions) were each observed from two directions as eight observation positions. The observation range is a region of 1,500 $\mu\text{m}\times$ 1,500 μm per site.

[0076] Each of the five substrates was observed at the above-described eight sites, and the ratio [number of sites where cracks were observed $\times$ 100/40 sites] (unit: %) of the sites where cracks were observed among the total of 40 sites was defined as the crack occurrence rate in the liquid bath temperature cycle test.

Fracture Test by Nanoindenter

Examples 1-6

1. Preparation of Test Piece

[0077] FIG. **4** is a transparent perspective view for explaining the position where the test piece was sampled and the position where the fracture test was performed on the test piece.

[0078] Printed wiring board substrates **1** to **6** obtained in Production Examples 1 to 6 were cut, and test pieces each having a size of 1.5 mm (cutting width W) $\times$ 60 mm (substrate length L) $\times$ 1.7 to 2.0 mm (substrate thickness T) indicated by a portion **20** in FIG. **4** were sampled.

[0079] In the sampled test piece, the test region where the fracture test is performed is a region indicated by a region **30** in FIG. **4**. The region **30** is a cross sectional surface of an end portion of the printed wiring board substrate **1**, and is substantially a central portion of the printed wiring board substrate **1** in a cross sectional view. The distance (distance B in FIG. **4**) to the region **30** from the corner of the printed wiring board substrate **1** is about 10 mm.

[0080] The cross sectional surface of the printed wiring board substrate including the region **30** was subjected to mechanical polishing using polishing paper of No. 4000 in advance. The obtained test piece was heat-treated at 260°C . for 6 hours in an air atmosphere, then cooled to room

temperature, and subjected to measurement by a nanoindenter.

2. Fracture Test by Nanoindenter

[0081] The test piece obtained as described above was mounted on the nanoindenter so that the indenter of the nanoindenter was pushed into the test region of the test piece. Thereafter, the position of the indenter was adjusted by an optical microscope and a micrometer attached to the nanoindenter so that the shortest distance between the glass cloth and the center position on the resin cured product where the indenter was pushed into was 5 μm on the surface into which the indenter was pushed, and the fracture test was performed under the following conditions.

Conditions

[0082] Device: product name “DUH-211S” manufactured by SHIMADZU CORPORATION

[0083] Type of indenter: Berkovich indenter (with a ridge interval of 115°) [0084] Indentation

speed: 39 mN/s [0085] Maximum load: 490 mN [0086] Measurement temperature: 25°C . [0087]

Measurement atmosphere: in air

3. Analysis of Measurement Results

[0088] Among the crack occurrence peaks appearing on the load-contact stiffness curve obtained by the fracture test described above, a crack occurrence peak having a contact stiffness decrease rate of 37% or more as calculated by the above equation (1) was identified.

[0089] **S1** in the above equation (1) was determined by the following procedure. FIG. 5 is a schematic diagram for explaining the method of determining **S1**.

[0090] The contact stiffness that is the peak top of the peak to be analyzed is defined as **S2**, the load that gives **S2** is defined as $W_{\text{sub.S2}}$, $W_{\text{sub.S2}}-25\text{ mN}$ is defined as $W_{\text{sub.1A}}$, and $W_{\text{sub.S2}}-15\text{ mN}$ is defined as $W_{\text{sub.1B}}$. Next, in the range of $W_{\text{sub.1A}}$ to $W_{\text{sub.1B}}$, an approximate straight line of the load-contact stiffness curve is obtained by the least squares method, and the straight line is defined as “approximate straight line **1**”.

[0091] Next, the absolute value **D** of the difference between the contact stiffness **S2** and the contact stiffness of the approximate straight line **1** at $W_{\text{sub.S2}}$ is calculated, the load that gives the contact stiffness of $S2+0.7D$ is defined as $W_{\text{sub.2A}}$, the load that gives the contact stiffness of $S2+0.5D$ is defined as $W_{\text{sub.2B}}$, the approximate straight line of the load-contact stiffness curve in the range of $W_{\text{sub.2A}}$ to $W_{\text{sub.2B}}$ is obtained by the least squares method, and the straight line is defined as “approximate straight line **2**”.

[0092] The contact stiffness at the intersection of the approximate straight line **1** and the approximate straight line **2** obtained as described above was determined as **S1**.

[0093] Among the crack occurrence peaks having a contact stiffness decrease rate of 37% or more as identified by the above-described method, a crack occurrence peak appearing at the lowest load was identified, and the load giving the peak top of the crack occurrence peak was determined as crack occurrence load **W1**.

[0094] FIG. 6 shows a graph in which crack occurrence load **W1** obtained as described above is plotted on the horizontal axis and the crack occurrence rate after the liquid bath temperature cycle test (1,000 cycles) is plotted on the vertical axis, and a linear approximate straight line of the plots obtained by the least squares method.

[0095] As is clear from FIG. 6, there is a high correlation between crack occurrence load **W1** and the crack occurrence rate after the liquid bath temperature cycle test (1,000 cycles), and it is found that the reliability test method of the present embodiment is useful as a reliability test method for printed wiring board substrates.

REFERENCE SIGNS LIST

[0096] **1** Printed wiring board substrate [0097] **10** Observation region [0098] **20** Sampled portion of test piece [0099] **30** Test region [0100] **W** Width [0101] **L** Length [0102] **T** Thickness [0103] **B** Distance [0104] **S1**, **S2** Contact stiffness [0105] $W_{\text{sub.1A}}$, $W_{\text{sub.1B}}$, $W_{\text{sub.2A}}$, $W_{\text{sub.2B}}$, $W_{\text{sub.S2}}$ Weight [0106] **A** Region

Claims

1. A reliability test method for a printed wiring board substrate, the method comprising measuring a load at which a crack occurs in the printed wiring board substrate, using a nanoindenter.
 2. The reliability test method for a printed wiring board substrate according to claim 1, wherein the printed wiring board substrate includes a resin cured product and a glass cloth.
 3. The reliability test method for a printed wiring board substrate according to claim 1, wherein the printed wiring board substrate has two or more layers each including a resin cured product and a glass cloth.
 4. The reliability test method for a printed wiring board substrate according to claim 2, wherein an indenter of the nanoindenter is pushed into the resin cured product.
 5. The reliability test method for a printed wiring board substrate according to claim 1, wherein an indenter of the nanoindenter is pushed into a cross sectional surface of the printed wiring board substrate.
 6. The reliability test method for a printed wiring board substrate according to claim 1, wherein a Berkovich indenter is used as an indenter of the nanoindenter.
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